

Class 14 Intro to Groups

Goals:

1. Understand what is meant by algebraic structure.
2. Know definition of a group.
3. Become familiar with examples of groups.

Definition: An algebraic structure is

Definition: A group is

1. Closure:

2. Associativity:

3. Identity:

4. Inverse:

Examples:

Why do the following sets fail to form a group under the given operation?

Integers under subtraction:

$\mathbb{Z}_n$ under regular addition:

Rational numbers under multiplication:

Irrational numbers under addition:

Three properties every group has:

Unique Identity Theorem: The identity element in any group is unique.

Unique Inverse Theorem: In any group, G , the inverse of any element, a , will be unique.

Left Cancellation Theorem: For any elements a , b , and c in a group, G , if $a \circ b = a \circ c$ then $b = c$.

Proofs

1. Every group has a unique identity element.

2. If a is any element of G then a has a unique inverse element.

Generally denote the unique inverse of a by a^{-1} . If the operation is addition we denote the inverse of a by $-a$.

3. If a , b , and c are any elements of a group and $a \circ b = a \circ c$, then $b = c$.